CHANDU PRASAD BHAIRAPU



Backend Software Engineer • AI Systems (Python, RAG, Agent Workflows)

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ABOUT ME

Backend engineer for AI systems with an MSc in Artificial Intelligence and 3 years of professional experience, writing Python backends for RAG and agent workflows and deploying them (AWS, Azure). At an early-stage startup I owned a knowledge-extraction pipeline end-to-end (ingestion, retrieval, inference), raised its precision/recall to 93%/90%, and moved a high-volume stream from LLM calls to deterministic code to cut cost. I build APIs with FastAPI and Python microservices and deploy with Docker, Kubernetes, AWS CDK, Azure and GitLab CI/CD.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning the knowledge-extraction pipeline end-to-end (data ingestion, chunking, prompt engineering, and indexing with agentic, deterministic, and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Raised precision/recall from 33%/45% to 93%/90% (measured against SAP-system-derived ground truth) with a multi-agent LangGraph extraction workflow: entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel over complex regulatory and technical documents.
- Built a deterministic (code-based) extraction pipeline instead of LLMs for a second, high-volume source stream, trading LLM cost for code: cut a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) query service with millisecond-latency retrieval from the knowledge graph (GraphRAG), integrated into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (Python microservices backend, React frontend) deployed on Azure, so domain experts validate and approve AI-extracted knowledge-graph changes.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, RAG, GraphRAG, Vector Search, MCP, FastAPI, Neo4j, MongoDB, Postgres, Kubernetes, Docker, Azure, React**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an agentic AI application with LangGraph that routes each request to vector database retrieval, general inference, or summarization, taking it from prototype to deployment on AWS.
- Built the Python backend and REST APIs with FastAPI (tested with Postman) and a React frontend; containerized the stack with Docker and deployed it to AWS using CDK, with GitLab CI/CD.
- **Tech stack: Python, FastAPI, LangGraph, LangChain, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, Postman, React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Designed RAG pipelines and compared chunking strategies, vector databases, indexing and fine-tuned retrieval models.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data to compare retrieval strategies.
- Automated evaluation of RAG generation and retrieval with RAGAS and PytrecEval, scoring model answers against chemist-provided ground truth.
- Built a scalable backend on event-driven AWS infrastructure (Lambda, SQS, DynamoDB) with CDK for the PrismaChat RAG evaluation.

- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, Vector Databases, Azure AI Search, AWS (Lambda, SQS, DynamoDB, S3, CDK), Docker, PyTorch**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data, served to chemists through a Streamlit dashboard.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost, FB Prophet, and LSTM models for EV load forecasting; the LSTM reached 96% test accuracy.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Autohaus Service Agent | Python | FastAPI | LangGraph | Qdrant | Postgres

- Python backend for a dealership service assistant: ingestion of synthetic workshop manuals and price lists into Qdrant embeddings, dense + keyword retrieval with context management (token budget), and a LangGraph agent that drafts appointment and cost-estimate answers with state persisted in Postgres, retries, and human handoff on tool errors.
- Evaluation and monitoring: a labelled question set scores retrieval recall and answer groundedness (hallucinations, RAGAS) in CI; every request logs latency, tokens, and cost, and a weekly comparison of eval scores flags drift.

Project: LLM Gateway | Python | FastAPI | LLM APIs

- Provider-agnostic API layer over multiple LLM provider APIs with token budgets, streaming, timeouts, retries, and fallback between providers, logging latency, token usage, and cost for every request.
- Metrics endpoint for cost per request, latency percentiles, and eval pass rate; tests and eval run in CI.

TECHNICAL SKILLSET

- **Backend & APIs:** Python, FastAPI, REST APIs, Python microservices, system architecture for AI pipelines, SQL, Postman, React (basic)
- **AI Systems:** AI pipelines (ingestion, chunking, retrieval, inference), RAG, GraphRAG, embeddings, vector search, retrieval evaluation, LangGraph (agent workflows, workflow orchestration, multi-agent), LangChain, LiteLLM, MCP, prompt engineering, evaluation (RAGAS, PytrecEval), Hugging Face Transformers
- **Databases:** Postgres, MongoDB, Neo4j, DynamoDB, vector databases (Qdrant, Pinecone, Azure AI Search)
- **Cloud & DevOps:** Docker, Kubernetes, AWS (CDK, Lambda, SQS, EC2, S3, VPC), Azure, GitLab CI/CD, Git
- **AI Development Tools:** Claude Code (custom skills, agentic coding workflows, MCP integration)
- **Machine Learning:** Scikit-Learn, PyTorch, TensorFlow, XGBoost, time-series analysis

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering, Minor in Computer Science & Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles (conference publication)

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)